Online Load Balancing under Stochastic Arrivals

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Introduction. Consider the following fundamental problem with broad applications: n machines receive a sequence of m jobs arriving online. Each job j has processing times p_{ji} on machine i and must be assigned immediately upon arrival to one of the machines, without future knowledge or the possibility of reassignment. The goal is to minimize the maximum load on any machine. This problem, known as the *online load balancing* problem, is central to algorithmic theory, particularly online algorithms, with far-reaching implications for practical systems such as *task schedulers* in operating systems, *resource allocators* in cloud computing, and *request handlers* in distributed systems.

When job arrivals are adversarial, a classical result [ABK92] establishes that the greedy algorithm achieves an $O(\log n)$-competitive ratio. Recall that an online algorithm is α-competitive if the ratio between its expected performance and the expected offline optimum is at most α. This upper bound is tight: even in the restricted *online graph balancing* setting – where edges of an unknown graph arrive online and must be immediately assigned to one endpoint to minimize the maximum in-degree – the competitive ratio cannot be improved beyond $\Omega(\log n)$. Notice that the graph balancing setting corresponds to the special case where each job can be assigned to only two machines, each with unit processing time.

Stochastic Arrivals. In many real-world settings, however, inputs are not truly adversarial. For instance, one could use historical data to learn arrival distributions. A standard way to mathematically model this in theoretical computer science is to assume that inputs arrive i.i.d. from a known fixed distribution $\mathcal{D}$. In many prior online problems such as the secretary [HKP04] and online Steiner tree [GGLS08] problems, exponential improvements in the competitive ratio have been achieved when shifting from the adversarial to the i.i.d. setting. Thus, we have the following conjecture that forms the motivation for this proposal:

Conjecture 1. *For the online load balancing problem, if jobs are drawn i.i.d. from a fixed known distribution $\mathcal{D}$, there exists an algorithm that achieves a $O(\mathsf{poly}(\log\log n))$ competitive ratio.*

Connection to Power-of-Two-Choices and Progress for Graphical Allocation. Even in the adversarial setting, the core difficulty of online load balancing appears in the restricted graph balancing problem. Therefore, a first step toward addressing this challenge is to study the *online graph balancing* setting where each edge is drawn i.i.d. uniformly from a fixed, known base graph G; this model is also known as *graphical allocation* [KP06]. When G is the complete graph K_n and there are n edge arrivals, the behavior of the greedy algorithm is well understood due to the influential *power-of-two-choices* result [ABKU94], which shows that the maximum in-degree is $O(\log\log n)$. This was later extended in [KP06] to regular graphs, where the authors showed that greedy remains $O(\log\log n)$-competitive. To the best of my knowledge, efforts into significantly generalizing the power-of-two-choices result beyond regular graphs have been limited, let alone the more general setting beyond graphs and with arbitrary processing times.

In my recent work with Sahil Singla, Milind Prabhu, and Nikhil Bansal (in submission), we take preliminary steps towards resolving Conjecture 1, by generalizing the power-of-two-choices result to general graphs:

Theorem 2. *For any arbitrary graph G on n vertices, there is an $O(\log\log n)$-competitive algorithm for graphical allocation on G.*

Two fundamental issues make this already restricted problem significantly more challenging than non-stochastic settings: (i) the expected offline optimum is a complicated random variable, and (ii) the greedy algorithm is suboptimal, and suffers a $\Omega(\log n/(\log\log n)^2)$ lower bound in the competitive ratio, despite its vast success in most previously considered settings. The existence of a positive result is promising for this direction of research, and its non-trivial nature makes the overarching question technically appealing, as it could lead to several new, useful tools in the design of online algorithms.

Next Steps. Given the nuances of the problem even when restricted to the graphical allocation setting, completely resolving Conjecture 1 will take a significant number of incremental efforts. There are two fundamental restrictive assumptions in the graphical allocation setting that must be relaxed to make progress towards resolving the conjecture: (i) machines are *restricted*, i.e., each job is restricted to a subset of two machines on which it can run (ii) machines are *identical*, i.e., each job has the same processing time on its two eligible machines, while it has infinite processing times on all other machines. I plan to push past these assumptions incrementally with the following thrusts:

(i) **Hypergraphical Allocation.** A natural extension of our work that relaxes the *restricted* machine assumption is to move from graphical to *hypergraphical allocation*, where each hyperedge is drawn i.i.d. uniformly at random from a fixed hypergraph and must be oriented to one of its endpoints. I am continuing to collaborate with the same team on this more general setting, and we are currently working to resolve the conjecture for k-uniform hypergraphs. Notably, several of our techniques from the graphical allocation setting fail when $k = \omega(1)$. To address this, I plan to approach the problem by separately considering:

1. $k = O(1)$: Our preliminary results suggest that more intricate behavior arises even in the $O(1)$-uniform hypergraphical case. However, there is room to adapt our techniques for hypergraphs; in particular, our "witness tree" proof approach for the load upper bound and the graph decomposition have elegant generalizations for the k-uniform hypergraphical setting, to yield a $O(\mathsf{poly}(\log\log n))$-competitive algorithm.

2. $k = \Omega(\log n)$: The most notable prior work in this regime is Godfrey's [God08] elegant result and its strengthening in [BBFN10], that if the k-uniform hypergraph is nearly-regular, then the maximum load of any node is at most $O(1)$. The simplified analysis in [BBFN10] involves a simple but smart probabilistic argument, which can be adapted to show that k-uniform hypergraphs satisfying a certain "bipartite" structure with bounded degrees also achieve $O(1)$ load. However, a decomposition of an arbitrary hypergraph into such pieces seems difficult.

The regime of $k = \omega(1)$ but $k = O(\log n)$ would likely be the most challenging, and I plan to tackle that setting the last by using an interpolation of the ideas from the two regimes above.

(ii) **Unrelated Online Graph Balancing.** To push past the assumption that machines are *identical*, a compelling next case to consider is when each job has a finite processing time on at most two machines, which can be modeled as a graph where each edge has different weights at its endpoints. Call this the *unrelated online graph balancing* problem. To tackle this, I will revisit the expected offline optimum, since insights from the identical case do not directly carry over. In the most general setting, understanding the offline optimum typically involves a convex program [AZM18]. I plan to study if further properties of the optimal solution can be uncovered in the structured two-machine setting.

Further, in the related *learning-augmented* setting of the online load balancing problem, recent works by Lattanzi et. al. [LLMV] and Shi Li and Jiayi Xian [LX21], have shown that when given access to predictions of the optimal solution to the convex program, there exists an online rounding algorithm that achieves an $O(\log\eta \cdot \log\log n/\log\log\log n)$, where η denotes the multiplicative error in the predictions. I aim to use the distributional information in the structured two-machine setting to obtain $O(\mathsf{poly}(\log n))$-approximate optimal solutions, which would then imply an $O(\mathsf{poly}(\log\log n))$-competitive algorithm.

I strongly believe these directions represent concrete and tractable steps toward the broader research goal, grounded in the momentum and insights developed so far.

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